Robust In-Car Child Presence Detection using Commercial WiFi

Sakila S. Jayaweera*, Beibei Wang†, K. J. Ray Liu†
†Origin Research, Rockville, USA, *University of Maryland, College Park, USA

Abstract

In-car child presence detection (CPD) has become a crucial vehicular safety measure to prevent heat-related injuries or fatalities of unattended children. However, existing CPD solutions based on sensors and cameras have limitations in accuracy, coverage, and additional cost. To address these challenges, leveraging existing in-car WiFi facilities has emerged as a promising solution. Nevertheless, current WiFi-based CPD solutions are not robust against environmental impacts and often generate unnecessary detection alerts when an adult is present in the car. In this demonstration, we propose a novel, robust CPD architecture designed to enhance performance under various conditions while minimizing false alarms. Our design incorporates three main systems: presence detection, seat localization, and child vs. adult classification. To validate the system's performance, we developed a prototype using existing WiFi devices operating at 5 GHz and conducted a series of experiments. Please find the project page with the URL: https://sites.google.com/originwirelessai.com/in-car-cpd.

CCS Concepts

• **Human-centered computing → Ubiquitous and mobile computing systems and tools**.

Keywords

Child presence detection(CPD), WiFi sensing, in-car sensing

ACM Reference Format:
Sakila S. Jayaweera*, Beibei Wang†, K. J. Ray Liu†. 2024. Robust In-Car Child Presence Detection using Commercial WiFi . In *The 30th Annual International Conference on Mobile Computing and Networking (ACM MobiCom '24), November 18–22, 2024, Washington D.C., DC, USA.* ACM, New York, NY, USA, 3 pages. https://doi.org/10.1145/3636534.3698864

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivs International 4.0 License.
ACM MobiCom '24, November 18–22, 2024, Washington D.C., DC, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0489-5/24/11
https://doi.org/10.1145/3636534.3698864

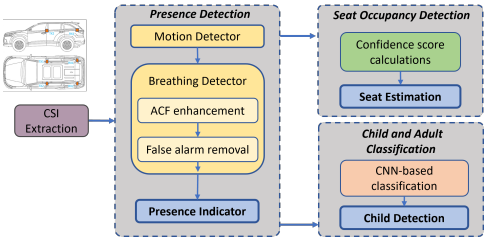


Figure 1: CPD system design.

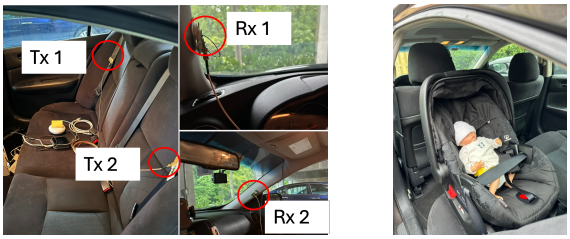
1 Introduction

With the increasing number of child heatstroke deaths reported over the past decades, CPD has become an essential vehicular technology for next-generation automobiles. However, earlier CPD solutions based on physical sensors [1] and cameras [2] either struggled to maintain higher accuracy due to the limited coverage, privacy-invasive, or required additional hardware for deployment. While RF-based solutions gain popularity for their privacy-preserving capabilities [7, 8], mmWave has limited coverage due to the field of view and is not readily available in many vehicles.

In contrast, WiFi-based solutions [5, 9, 10] provide extensive coverage, better privacy, and easy installation with no additional cost, as many new vehicles are already equipped with in-car WiFi facilities. However, existing solutions are not robust under unfavorable environmental conditions and cannot differentiate child presence from adult presence [3, 11, 12].

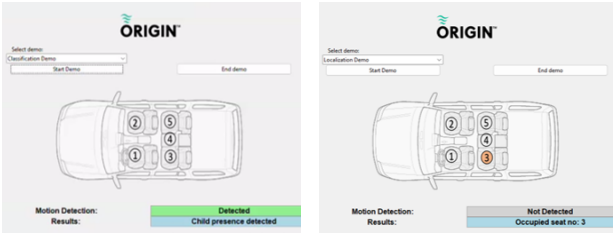
In this demonstration, we propose a robust CPD system with three separate modules: presence detection, seat localization, and child vs. adult classification to address the aforementioned challenges. Fig. 1 illustrates the system design. The presence detection module utilizes image enhancement on the breathing spectrum to improve the detection of subtle breathing movements, which are often corrupted by environmental impacts. The seat localization module estimates the occupied seat based on a confidence score calculated from the breathing spectrum. The child vs. adult classification module uses a CNN-based neural network architecture to extract the necessary features for distinguishing between the presence of children, adults, and empty scenarios.

Extensive experiments across 25 car models validate the system's performance in real-time scenarios. Additionally, Murata has commercialized Japan's first in-vehicle child presence detection system using our WiFi sensing technology, demonstrating the feasibility of real-world deployment [6].



(a)Antenna placement (b) Baby doll

Figure 2: CPD demonstration hardware setup and baby doll placement with a baby seat inside the car.



(a) Classification demo (b) Localization demo

Figure 3: WebGUI design for the demo.

2 System Design

The proposed CPD system consists of three separate modules.

2.1 Presence Detection

The presence detection module detects the presence of life inside the car regardless of whether the person is awake or in a stationary position, such as sleep. To detect the presence, the module utilizes two detectors: a motion detector and a breathing detector.

The motion detector identifies human movements through motion statistics, which are extracted from the CSI power's autocorrelation function (ACF) [4, 13, 14]. In the absence of motion but breathing, ACF demonstrated periodic patterns corresponding to the breathing signal [15]. Under unfavorable environmental conditions, the periodic pattern in ACF can be corrupted. In such scenarios, we apply peak enhancement based on the first derivative of the signal, making peaks more distinct and easier to identify. Uneven contrast caused by the peak enhancement step is removed using histogram equalization. We utilize a two-metric zero crossing rate and dynamic time warping to remove false detection.

While the above method improves the overall performance, we observed occasional miss detections and sporadic false detections. We utilize window-based decision fusion to smooth out the final detection and eliminate these sudden changes.

2.2 Seat Localization

We utilize a confidence score-based detection method that uses the ACF extracted from each Tx-Rx pair. Here, we assume that the pair of transmitter (Tx) and receiver (Rx) antennas are placed separately in four corners inside the vehicle. An example antenna setup is depicted in Fig.2(a), where two Rx antennas are placed near two side mirrors and two Tx antennas at the two corners of the rear seat. We further assume that there is only one occupant inside the car and maintain a static position to extract breathing for a certain time window. We analyze the ACF extracted from each link and calculate the confidence score for each ACF. The proximity of a user to the antenna influences the confidence score. For instance, if the target is in the rear seat as depicted in Fig.2 (b), ACF corresponding to Tx 2- Rx 1 and Tx 2- Rx 2 shows higher confidence scores compared to the other two pairs. Given the result, we expect a high probability that the target subject is near the Tx 2, the driver-side rear seat. In contrast to the other four seats, the rear middle seat demonstrates similar levels of confidence scores for all four links.

2.3 Child vs. Adult Classification

Even though children and adults are easily categorized based on breathing rates, moving children cannot be differentiated from adults using motion statistics. As a result, we infer that ACF does not have the information required for such classification. Therefore, we utilize CSI and CNN-based network architecture to extract the relevant features. Here, we utilize 4 CNN layers followed by max-pooling layers. After extracting features using the CNN layers, two fully connected layers are used to classify the feature into a correct class. We propose a lightweight CNN model as we need to minimize the complexity when implementing it on the edge.

3 Demonstration Setup

Fig. 2 illustrates the hardware setup, which utilizes a commercial WiFi chipset operating at 5 GHz with a bandwidth of 40 MHz. Each chipset includes two antennas that can be positioned in the four corners of the vehicle. To obtain the CSI, we modified the WiFi chipset driver. The estimated CSI comprises 58 subcarriers with a 30Hz sampling rate.

We demonstrate the human presence detection utilizing the demo kit which shows the live motion, breathing rates and presence results. The Web GUI, shown in Fig. 3, displays the CPD detection results in real-time for classification and localization demos, and video of demos can be found in our project website. From experiments conducted across 25 different car models, our system achieved a 99.8% detection rate with only a 1.1% false alarm rate using the presence detection module. Additionally, the system demonstrated 92.44% overall accuracy for seat occupancy detection and 95.86% accuracy for CNN-based classification.

References

[1] N. Hashim, H. Basri, A. Jaafar, M. Aziz, A. Salleh, and A. S. Ja. 2014. Child in car alarm system using various sensors. *ARPN Journal of Engineering and Applied Sciences* 9, 9 (2014), 1653–1658.

[2] J. Jaworek-Korjakowska, A. Kostuch, and P. Skruch. 2021. SafeSO: Interpretable and Explainable Deep Learning Approach for Seat Occupancy Classification in Vehicle Interior. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. 103–112. https://doi.org/10.1109/CVPRW53098.2021.00020

[3] Sakila S. Jayaweera, Beibei Wang, Xiaolu Zeng, WeiHsiang Wang, and K. J. R Liu. 2023. WIFI-Based Robust Child Presence Detection for Smart Cars. In *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 1–5. https://doi.org/10.1109/ICASSP49357.2023.10096908

[4] K. J. R. Liu and B. Wang. 2019. *Wireless AI: Wireless Sensing, Positioning, IoT, and Communications*. Cambridge University Press. https://doi.org/10.1017/9781108597234

[5] K. J. R. Liu and Beibei Wang. 2024. Statistical Principles of Time Reversal [Perspectives]. *IEEE Signal Processing Magazine* 41, 1 (2024), 31–37. https://doi.org/10.1109/MSP.2023.3324472

[6] Murata Manufacturing Co. Ltd. 2024. Murata Commercializes Japan's First In-Vehicle Child Presence Detection System Using Wi-Fi Sensing. https://corporate.murata.com/en-us/newsroom/news/company/general/2024/0604 Accessed: 2024-09-06.

[7] Kotone Sato, Steven Wandale, Koichi Ichige, Kazuya Kimura, and Ryo Sugiura. 2024. Infant Detection Using In-Cabin Millimeter- Wave FMCW-MIMO Radar and CFAR Algorithm. In *2024 International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC)*. 1–6. https://doi.org/10.1109/ITC-CSCC62988.2024.10628229

[8] Kotone Sato, Steven Wandale, Koichi Ichige, Kazuya Kimura, and Ryo Sugiura. 2024. Millimeter-Wave Radar-Based Vehicle In-Cabin Occupancy Detection Using Explainable Machine Learning. *IEEE Sensors Journal* 24, 15 (2024), 24288–24298. https://doi.org/10.1109/JSEN.2024.3413775

[9] B. Wang, Q. Xu, C. Chen, F. Zhang, and K. J. R. Liu. 2018. The Promise of Radio Analytics: A Future Paradigm of Wireless Positioning, Tracking, and Sensing. *IEEE Signal Processing Magazine* 35, 3 (2018), 59–80. https://doi.org/10.1109/MSP.2018.2806300

[10] Chenshu Wu, Beibei Wang, Oscar C. Au, and K. J. R. Liu. 2022. Wi-Fi Can Do More: Toward Ubiquitous Wireless Sensing. *IEEE Communications Standards Magazine* 6, 2 (2022), 42–49. https://doi.org/10.1109/MCOMSTD.0001.2100111

[11] Q. Xu, B. Wang, F. Zhang, S. D. Regani, F. Wang, and K. J. R. Liu. 2020. Wireless AI in Smart Car: How Smart a Car Can Be? *IEEE Access* 8 (2020), 55091–55112. https://doi.org/10.1109/ACCESS.2020.2978531

[12] X. Zeng, B. Wang, C. Wu, S. D. Regani, and K. J. R. Liu. 2022. WiCPD: Wireless Child Presence Detection System for Smart Cars. *IEEE Internet of Things Journal* (2022).

[13] F. Zhang, C. Chen, B. Wang, and K. J. R. Liu. 2018. WiSpeed: A Statistical Electromagnetic Approach for Device-Free Indoor Speed Estimation. *IEEE Internet of Things Journal* 5, 3 (2018), 2163–2177. https://doi.org/10.1109/JIOT.2018.2826227

[14] F. Zhang, C. Wu, B. Wang, H. Q. Lai, Y. Han, and K. J. R. Liu. 2019. WiDetect: Robust motion detection with a statistical electromagnetic model. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 3, 3 (2019), 1–24.

[15] F. Zhang, C. Wu, B. Wang, M. Wu, D. Bugos, H. Zhang, and K. J. R. Liu. 2019. SMARS: Sleep monitoring via ambient radio signals. *IEEE Transactions on Mobile Computing* 20, 1 (2019), 217–231.